Comprehensive Operational Roadmap, Hardware Specifications, Data Mining Protocols, and Task Distribution for Multiscale Dynamic Colloid Simulation

Short Title: Operational Roadmap for Dynamic Colloid Simulation

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Abstract

Bridging microscopic surfactant thermodynamics with macroscopic continuum fluid dynamics requires a structured computational infrastructure, precise experimental validation datasets, and an efficient division of labor between specialized artificial intelligence agents. In this operational roadmap, we delineate the complete technical requirements and execution blueprint for the Surfactant-Dynamics-Sim project across four sequential phases. First, we outline the exact data mining requirements, focusing on four primary classes of experimental records: dynamic surface tensiometry (DST), large-amplitude oscillatory dilatational (LAOD) interfacial rheology, microfluidic droplet pinch-off timeseries, and thin liquid film foam drainage isotherms. Second, hardware and software environments are comprehensively specified, detailing the utilization of local Intel Core Ultra 7 CPUs, 8 GB Intel Arc 140V GPUs (for neural operator surrogate acceleration), remote Kaggle GPU environments (for heavy coarse-grained molecular dynamics and Bayesian sampling), and the pre-installed Scilab-Xcos 2025 numerical engine. Third, an operational collaboration framework is established between Antigravity (the local execution and numerical agent) and Arena AI (the browser-based data mining and literature extraction agent), synchronized through GitHub (prashant-kotian/surfactant-dynamics-sim) and Google Drive (kotianinterface@gmail.com). Finally, a rigorous four-phase plan of action is detailed, providing milestone criteria, numerical convergence tolerances, and journal submission targets for high-impact Q1 publications.

Keywords: SurfactantKit, Scilab, Xcos, data mining, Intel Arc GPU, Kaggle GPU, agentic collaboration, interfacial dynamics.

1. Introduction and Objectives

The primary objective of the Surfactant-Dynamics-Sim initiative is to establish the first open-source, multiscale simulation framework coupling molecular-level surfactant thermodynamic properties with continuum-scale dynamic transport phenomena [1,2]. While standard molecular modeling toolkits and computational fluid dynamics packages operate in mutual isolation, our integrated framework bridges this gap by marrying the molecular parameter calculation engine of SurfactantKit with the stiff differential-algebraic equation (DAE) solvers of Scilab-Xcos [3-5].

To ensure seamless execution, this document serves as the master operational plan. It provides complete specifications for data acquisition, hardware utilization across local and cloud tiers, AI agent task allocations, and a phase-by-phase implementation schedule designed to yield premium scientific publications in premier journals such as Journal of Colloid and Interface Science (JCIS), Chemical Engineering Science (CES), and Colloids and Surfaces A [6-8].

2. Experimental Data Mining Protocols and Target Datasets

To rigorously validate the computational models across all dynamic regimes, experimental datasets must be extracted, digitized, and curated from peer-reviewed literature into standardized JSON and CSV formats [9]. Four core experimental categories are required:

2.1. Dynamic Surface Tensiometry (DST) Datasets

Dynamic surface tension curves (gamma(t) versus time) spanning short adsorption times (1 millisecond) to long-term equilibrium (10^3 seconds) are required for baseline model calibration [10,11]. Target surfactant systems must encompass diverse molecular architectures: (1) anionic surfactants, specifically sodium dodecyl sulfate (SDS) and sodium bis(2-ethylhexyl) sulfosuccinate (AOT); (2)

cationic surfactants, including cetyltrimethylammonium bromide (CTAB) and dodecyltrimethylammonium bromide (DTAB); (3) nonionic ethoxylated alcohols, such as Triton X-100 and C12E8; and (4) representative cationic gemini surfactants (alkanediyl-alpha,omega-bis(dimethylalkylammonium bromide), 12-s-12 series) [12,13]. For each system, digitized records must capture temperature, bulk concentration (from 0.01*CMC to 10*CMC), maximum surface excess (Gamma_inf), molecular headgroup area (a0), and diffusion coefficient (D).

2.2. Large-Amplitude Oscillatory Dilatational (LAOD) Rheology Datasets

Interfacial dilatational rheometry records obtained via pulsating bubble or oscillating pendant drop tensiometers (e.g., Kruss PAT-1M, Teclis Tracker) provide complex viscoelastic moduli as functions of angular frequency (omega = 0.05 to 10 rad/s) and strain amplitude (epsilon = Delta A / A0 = 0.01 to 0.40) [14,15]. Mined data must include the storage dilatational modulus (E'), loss modulus (E''), loss angle (theta), and higher harmonic intensity ratios (I_2omega / I_omega, I_3omega / I_omega) to benchmark non-linear viscoelastic distortions [16].

2.3. Microfluidic Droplet Breakup and Pinch-Off Timeseries

High-speed microphotographic datasets of droplet generation in planar flow-focusing and T-junction microchannels provide critical validation for dynamic interfacial area generation [17,18]. Extracted parameters must comprise channel orifice dimensions (width w = 20-100 micrometers), continuous and dispersed flow rates (Q_c, Q_d), dynamic capillary numbers (Ca = mu * v / gamma(t)), neck thinning trajectories (h_neck(t) versus time), and resulting droplet polydispersity indices.

2.4. Thin Liquid Film and Dynamic Foam Lamella Drainage

Thin film balance (TFB) interferometric measurements of microscopic foam lamellae (Scheludko cell data) are required to validate drainage hydrodynamics [19,20]. Target datasets comprise

film thickness decay curves (h(t) versus time), equilibrium film thickness (h_eq), critical rupture thickness (h_crit), and disjoining pressure isotherms (Pi(h)) across ionic strengths from 10^{-4} to 10^{-1} M.

Table 1. Summary of target experimental datasets for calibration and validation of dynamic continuum models.

Experimental Category	**Primary Technique**	**Key Physical Variables**	**Target Model Surfactants**	**Key Literature Sources**
Dynamic Surface Tension (DST)	Maximum bubble pressure / Pendant drop	gamma(t), Gamma_inf, D, ka, kd	SDS, CTAB, Triton X-100, 12-s-12	Miller et al. [10], Eastoe et al. [11]
Interfacial Dilatational Rheology	Oscillating drop tensiometry (PAT-1M)	E'(omega), E''(omega), theta, harmonics	SDS, C12E8, Beta-casein, Geminis	Lucassen et al. [14], Ravera et al. [15]
Microfluidic Droplet Pinch-Off	High-speed optical microfluidics	Q_c/Q_d, Ca(t), neck width, droplet size	AOT / mineral oil, SDS / hexadecane	Anna et al. [17], Garstecki et al. [18]
Thin Liquid Film Drainage	Thin film balance (TFB) interferometry	h(t), hcrit, disjoining pressure Pi(h)	SDS + NaCl, C12TAB, Alkyl polyglucosides	Exerowa et al. [19], Bergeron et al. [20]

3. Hardware Specifications and Computational Allocation

The computational architecture combines local edge processing with high-performance cloud accelerators to optimize turnaround time and numerical efficiency.

3.1. Local Host Environment: Windows 11 Laptop

The host workstation is equipped with an Intel Core Ultra 7 256V processor (8 high-IPC physical cores) and 16 GB on-package LPDDR5X memory. This environment hosts Scilab 2025.0.0 and Xcos, utilizing the Intel Math Kernel Library (MKL) for accelerated matrix decompositions and SUNDIALS CVode/IDA numerical integration [21]. All local Scilab scripts, ODE state-space solvers, and Xcos graphical block diagrams (.zcos) execute directly within this tier.

3.2. Intel Arc 140V GPU (8 GB) Utilization

The integrated Intel Arc 140V GPU, based on the advanced Xe2 microarchitecture with dedicated XMX AI engines, is allocated specifically for local machine learning acceleration [22]. While Scilab itself relies on CPU-bound MKL vectorization, the Intel Arc GPU is harnessed via PyTorch with

DirectML (torch-directml) or Intel Extension for PyTorch (IPEX) to train and execute Physics-Informed Neural Networks (PINNs) and DeepONet neural operators [23]. These neural surrogates approximate the non-local Ward-Tordai convolutional integral in sub-millisecond inference times, providing instant property evaluations to the Scilab solver.

3.3. Remote Cloud GPU Environment: Kaggle Compute

For tasks exceeding local workstation memory or requiring prolonged multi-GPU parallelism, a dedicated Kaggle cloud account (equipped with dual Nvidia Tesla T4 or P100/A100 accelerators) is utilized [24]. This environment is deployed for: (1) coarse-grained dissipative particle dynamics (DPD) or molecular dynamics (GROMACS) to compute structural micellar aggregation numbers; and (2) extensive Markov Chain Monte Carlo (MCMC) Bayesian parameter estimation sweeps across thousands of experimental DST curves. Job submissions, monitoring, and artifact downloads are managed programmatically via the Kaggle API.

3.4. Optional Linux Environment: Windows Subsystem for Linux (WSL2)

For headless Linux compilation, Modelica C-code generation, and automated regression testing pipelines, Windows Subsystem for Linux (WSL2 Ubuntu 24.04 LTS) is configured on the host machine. This tier ensures platform-independent reproducibility.

4. Agentic Collaboration and Division of Labor

To achieve maximum productivity, project workflows are partitioned between two specialized artificial intelligence agents: Antigravity (agy), operating locally on the host machine, and Arena AI, operating within a high-capability browser sandbox [25].

4.1. Role and Responsibilities of Antigravity (Local Execution Agent)

Antigravity functions as the core numerical and systems engineering agent. Its responsibilities comprise: (1) developing and maintaining Scilab scripts, Xcos block diagrams, and Python-Scilab

bridging routines; (2) compiling and running local simulations via the native Scilab engine; (3) managing the Intel Arc GPU for surrogate model training and running Kaggle GPU cloud jobs; (4) maintaining the GitHub repository (prashant-kotian/surfactant-dynamics-sim); and (5) authoring publication manuscripts strictly compliant with MANUSCRIPT_FORMATTING_SPEC.md.

4.2. Role and Responsibilities of Arena AI (Browser and Data Mining Agent)

Arena AI functions as the research intelligence and literature acquisition agent. Its responsibilities comprise: (1) querying academic databases (ScienceDirect, ACS, RSC, Springer) to locate target experimental studies; (2) extracting and digitizing graphical data curves from scanned PDFs using automated image processing and WebPlotDigitizer routines; (3) packaging digitized records into curated JSON schema files; and (4) uploading raw papers, literature syntheses, and digitized data tables to the shared Google Drive repository (kotianinterface@gmail.com) and creating pull requests on GitHub.

4.3. Synchronization Infrastructure

Bidirectional communication between Antigravity and Arena AI is established across two channels: (1) GitHub (prashant-kotian/surfactant-dynamics-sim) serves as the primary version-controlled codebase for models, scripts, and structured JSON datasets; and (2) Google Drive (kotianinterface@gmail.com) serves as the storage repository for large binary files, high-resolution microscope videos, raw literature PDFs, and intermediate simulation state files.

Table 2. Task partitioning and synchronization protocols between Antigravity and Arena AI.

Workflow Dimension	**Antigravity (Local Host Agent)**	**Arena AI (Browser Sandbox Agent)**	**Shared Synchronization Channel**
Literature Acquisition	Receives curated PDFs & schemas	Discovers, downloads & synthesizes papers	Google Drive / literature/ index
Data Digitization	Benchmarks digitized values in solvers	Extracts (x, y) coordinates from PDF figures	GitHub data/curated/ (JSON format)
Scilab-Xcos Modeling	Authors & executes .sce / .zcos files	Reviews mathematical equations & logic	GitHub scilab/ repository

GPU Acceleration	Intel Arc GPU (local) & Kaggle GPU (cloud)	Not applicable (browser sandbox only)	Kaggle API / output checkpoints
Manuscript Preparation	Generates .docx & verifies headless PDF	Contributes literature reviews & discussions	GitHub docs/ & Google Drive drafts

5. Detailed Four-Phase Plan of Action

The project is structured into four sequential, milestone-driven phases spanning an eight-month development lifecycle.

5.1. Phase 1: Analytical Verification and Baseline Solvers (Months 1-2)

Objective: Establish the foundational Scilab-Xcos numerical architecture and verify solver precision against exact analytical solutions.

Key Deliverables: (1) Construct Xcos state-space block diagrams for diffusion-controlled planar adsorption; (2) Benchmark SUNDIALS CVode solver against asymptotic short-time (gamma(t) proportional to $t^{0}.5$) and long-time (gamma(t) proportional to $t^{-0}.5$) Ward-Tordai solutions; (3) Verify linear Lucassen-van den Tempel (LVDT) viscoelastic limits for small strain amplitudes (epsilon < 0.02); and (4) Initialize the automated continuous integration pipeline on GitHub.

5.2. Phase 2: Experimental Data Mining and Model Validation (Months 3-4)

Objective: Ingest digitized experimental datasets from Arena AI and calibrate dynamic kinetic parameters across model surfactant classes.

Key Deliverables: (1) Arena AI curates 50+ benchmark DST and LAOD datasets into data/curated/; (2) Antigravity executes non-linear least-squares fitting in Scilab to extract kinetic rate constants (ka, kd) and micellar relaxation times (tau1, tau2); (3) Validate dynamic surface tension predictions against Kruss PAT-1M measurements for SDS, CTAB, and Triton X-100; and (4) Deploy Intel Arc GPU surrogate models for real-time Ward-Tordai convolution.

5.3. Phase 3: Non-Equilibrium Discovery and Parameter Sweeps (Months 5-6)

Objective: Explore unmapped physical regimes, identify novel non-linear phenomena, and construct predictive engineering regime maps.

Key Deliverables: (1) Map higher-order Fourier harmonic distortion (2*omega, 3*omega) under large-amplitude area oscillations (epsilon = 0.20-0.50); (2) Simulate dynamic Marangoni tip-streaming in microfluidic junctions, identifying critical Capillary numbers (Ca_crit) governing satellite droplet suppression; (3) Model acoustic resonance in thin liquid film drainage, predicting subharmonic rupture frequencies; and (4) Execute multi-GPU Bayesian sensitivity analyses on Kaggle.

5.4. Phase 4: Manuscript Preparation and Journal Submission (Months 7-8)

Objective: Package theoretical formulations, simulation codes, and experimental validations into peer-reviewed manuscripts.

Key Deliverables: (1) Author Manuscript 1 on Non-linear LAOD Rheology for Journal of Colloid and Interface Science (JCIS); (2) Author Manuscript 2 on Microfluidic Marangoni Droplet Breakup for Chemical Engineering Science (CES); (3) Publish open-source software release on GitHub with full documentation and interactive examples; and (4) Verify all manuscripts against MANUSCRIPT_FORMATTING_SPEC.md.

6. Conclusion and Immediate Next Steps

This operational roadmap provides an unambiguous, rigorous framework for transforming SurfactantKit and Scilab-Xcos into an internationally recognized computational platform for dynamic colloid science. By combining local Intel hardware and cloud GPU acceleration with a disciplined division of labor between Antigravity and Arena AI, the project is uniquely positioned to produce reproducible, high-impact scientific discoveries.

Immediate Action Items: (1) Clone the newly initialized GitHub repository (prashant-kotian/surfactant-dynamics-sim); (2) Establish the shared Google Drive directory

(kotianinterface@gmail.com); (3) Dispatch Arena AI to begin data mining of baseline DST literature for SDS, CTAB, and Triton X-100; and (4) Implement Phase 1 analytical Ward-Tordai benchmark tests in Scilab.

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